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DROP DATABASE IF EXISTS insurance;
CREATE DATABASE insurance;
USE insurance;

CREATE TABLE IF NOT EXISTS person (
driver_id VARCHAR(255) NOT NULL,
driver_name TEXT NOT NULL,
address TEXT NOT NULL,
PRIMARY KEY (driver_id)
);
CREATE TABLE IF NOT EXISTS car (
reg_no VARCHAR(255) NOT NULL,
model TEXT NOT NULL,
c_year INTEGER,
PRIMARY KEY (reg_no)
);
CREATE TABLE IF NOT EXISTS accident (
report_no INTEGER NOT NULL,
accident_date DATE,
location TEXT,
PRIMARY KEY (report_no)
);
CREATE TABLE IF NOT EXISTS owns (
driver_id VARCHAR(255) NOT NULL,
reg_no VARCHAR(255) NOT NULL,
FOREIGN KEY (driver_id) REFERENCES person(driver_id) ON DELETE CASCADE,
FOREIGN KEY (reg_no) REFERENCES car(reg_no) ON DELETE CASCADE
);
CREATE TABLE IF NOT EXISTS participated (
driver_id VARCHAR(255) NOT NULL,
reg_no VARCHAR(255) NOT NULL,
report_no INTEGER NOT NULL,
damage_amount FLOAT NOT NULL,
FOREIGN KEY (driver_id) REFERENCES person(driver_id) ON DELETE CASCADE,
FOREIGN KEY (reg_no) REFERENCES car(reg_no) ON DELETE CASCADE,
FOREIGN KEY (report_no) REFERENCES accident(report_no)
);
INSERT INTO person VALUES
("D111", "Driver_1", "Kuvempunagar, Mysuru"),
("D222", "Smith", "JP Nagar, Mysuru"),
("D333", "Driver_3", "Udaygiri, Mysuru"),
("D444", "Driver_4", "Rajivnagar, Mysuru"),
("D555", "Driver_5", "Vijayanagar, Mysore");
INSERT INTO car VALUES
("KA-20-AB-4223", "Swift", 2020),
("KA-20-BC-5674", "Mazda", 2017),
("KA-21-AC-5473", "Alto", 2015),
("KA-21-BD-4728", "Triber", 2019),
("KA-09-MA-1234", "Tiago", 2018);
INSERT INTO accident VALUES
(43627, "2020-04-05", "Nazarbad, Mysuru"),
(56345, "2019-12-16", "Gokulam, Mysuru"),
(63744, "2020-05-14", "Vijaynagar, Mysuru"),
(54634, "2019-08-30", "Kuvempunagar, Mysuru"),
(65738, "2021-01-21", "JSS Layout, Mysuru"),
(66666, "2021-01-21", "JSS Layout, Mysuru");
INSERT INTO owns VALUES
("D111", "KA-20-AB-4223"),
("D222", "KA-20-BC-5674"),
("D333", "KA-21-AC-5473"),
("D444", "KA-21-BD-4728"),
("D222", "KA-09-MA-1234");
INSERT INTO participated VALUES
("D111", "KA-20-AB-4223", 43627, 20000),

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("D222", "KA-20-BC-5674", 56345, 49500),
("D333", "KA-21-AC-5473", 63744, 15000),
("D444", "KA-21-BD-4728", 54634, 5000),
("D222", "KA-09-MA-1234", 65738, 25000);
```

```
SELECT * FROM person;
SELECT * FROM car;
SELECT * FROM accident;
SELECT * FROM owns;
SELECT * FROM participated;
```

```
-- Find the total number of people who owned a car that were involved in
accidents in 2021
select COUNT(driver_id)
from participated p, accident a
where p.report_no=a.report_no and a.accident_date like "2021%";
```

```
-- Find the number of accident in which cars belonging to smith were involved
select COUNT(distinct a.report_no)
from accident a
where exists
(select * from person p, participated ptd where p.driver_id=ptd.driver_id and
p.driver_name="Smith" and a.report_no=ptd.report_no);
```

```
-- Add a new accident to the database
insert into accident values
(45562, "2024-04-05", "Mandya");
SELECT * FROM accident;
```

```
insert into participated values
("D222", "KA-21-BD-4728", 45562, 50000);
select * from participated;
```

```
-- Delete the Mazda belonging to Smith
delete from car
where model="Mazda" and reg_no in
(select car.reg_no from person p, owns o where p.driver_id=o.driver_id and
o.reg_no=car.reg_no and p.driver_name="Smith");
select * from car;
```

```
-- Update the damage amount for the car with reg_no of KA-09-MA-1234 in the
accident with
report_no 65738
update participated set damage_amount=10000 where report_no=65738 and
reg_no="KA-09
MA-1234";
select * from participated;
```

```
-- View that shows models and years of car that are involved in accident
create view CarsInAccident as
select distinct model, c_year
from car c, participated p
where c.reg_no=p.reg_no;
select * from CarsInAccident;
```

```
-- A trigger that prevents a driver from participating in more than 2 accidents
in a given year.
```

```
DELIMITER //
create trigger PreventParticipation
before insert on participated
for each row
BEGIN
IF 2<=(select count(*) from participated where driver_id=new.driver_id) THEN
signal sqlstate '45000' set message_text='Driver has already participated in 2
accidents';
END IF;
END;//
DELIMITER ; //
INSERT INTO participated VALUES
("D333", "KA-20-AB-4223", 66666, 20000);
ERROR 1644 (45000): Driver has already participated in 2
accidents
```